

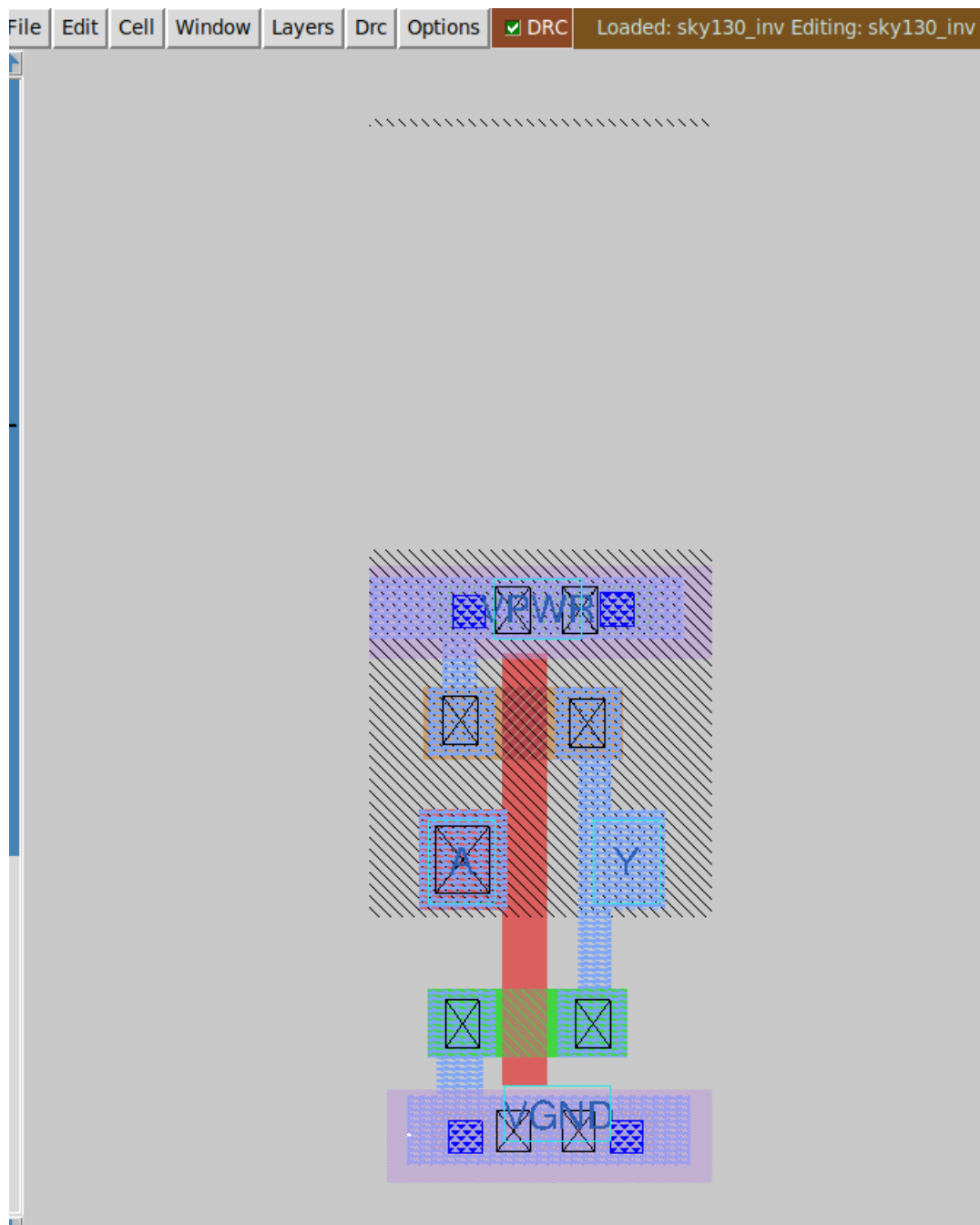
Lab introduction to Sky130 basic layers layout and LEF using inverter

This lab guides you in constructing and verifying the layout of a **basic CMOS inverter** using standard VLSI design tools. You will learn how to identify and use the necessary process layers (local interconnect, Metal1, Metal2, etc.), select the correct diffusion regions for NMOS and PMOS devices, verify device presence and connectivity using layout tools, and understand the difference between physical layout and abstract cell representations (like LEF). The lab steps walk through detailed operations for selection, connectivity checks, and an explanation of the LEF abstraction.

Lab Steps

1. Recognize and Identify CMOS Inverter Layers

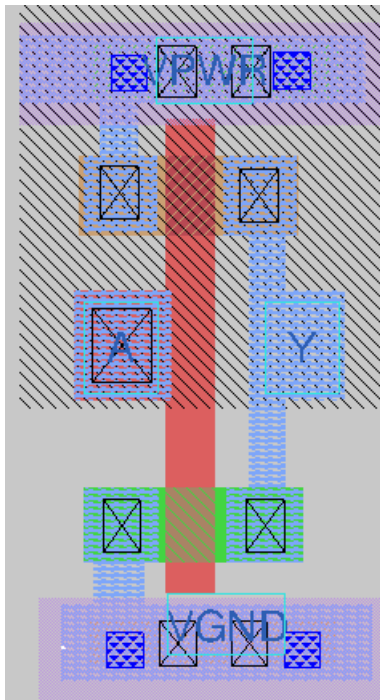
- **Objective:** Understand and identify each layer required for CMOS inverter layout in the layout editor (e.g., magic or SkyWater).
 - The **local interconnect layer** (*local I*) is used for short routing.
 - **Metal1** (light purple) and **Metal2** (pinkish) are upper-level routing layers.
 - The "**anvil**" is represented by solid, slanting dashed lines.
 - **NMOS diffusion** is marked in green; **PMOS diffusion** in brown.
 - **Polysilicon** (often red) forms the gate.
 - Use the color palette within the tool to identify each layer by hovering the cursor and observing the legend.



2. Confirm Diffusion Regions and Polysilicon Intersections

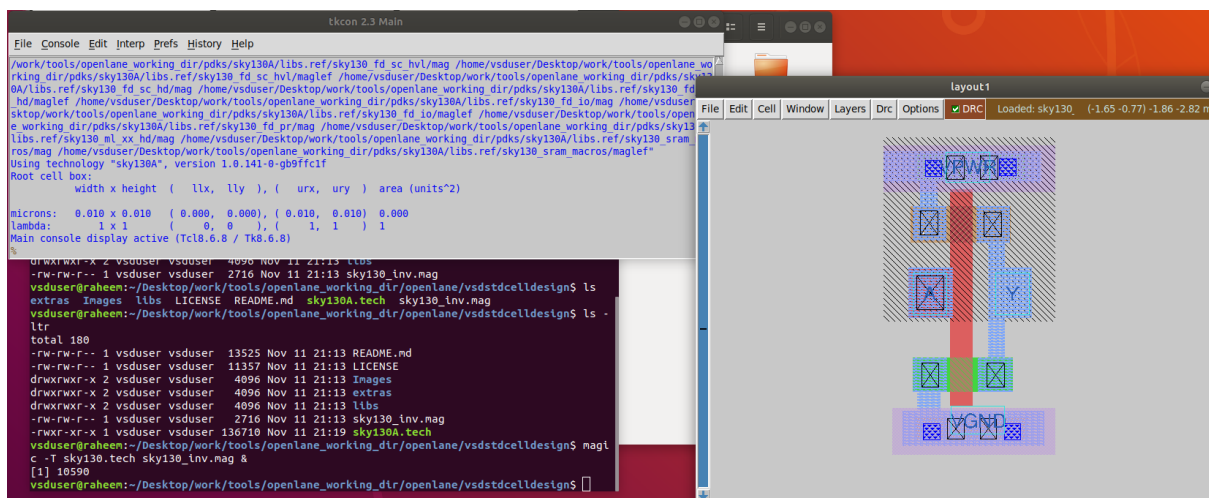
- **Objective:** Ensure devices are correctly instantiated according to CMOS design rules.
 - Where **polysilicon crosses NMOS diffusion (green)**, an **NMOS transistor** is formed.
 - Where **polysilicon crosses PMOS diffusion (brown)**, a **PMOS transistor** is formed.

- Hover over the intersection area in the layout tool and press **s** to select the object beneath the cursor.



Verify Device Structures

- With the area selected (after pressing **s**), use the tool's "what" function (sometimes invoked by a command or menu) to identify the selected structure.
 - Confirm that the intersection of **poly** and **n-diffusion** reports as **NMOS**.
 - Confirm that the intersection of **poly** and **p-diffusion** reports as **PMOS**.

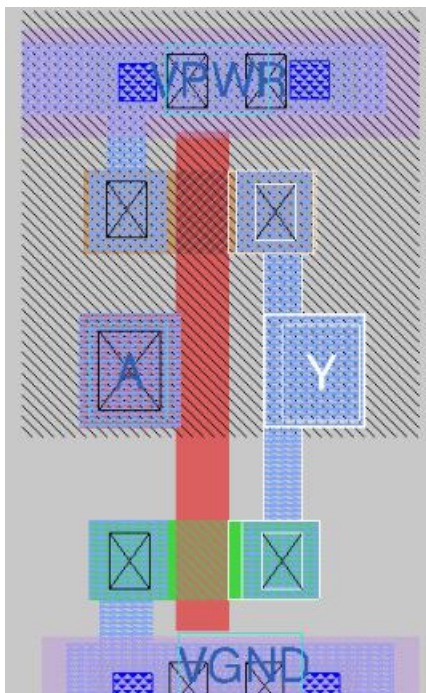


Examine Gate Connections

- **Objective:** Verify both gates are connected to the inverter input.
 - The gates of both NMOS and PMOS are connected to the input net.
 - Select the gate poly line and observe that both transistor gates are electrically connected.

Check Output Connectivity (Drain Connections)

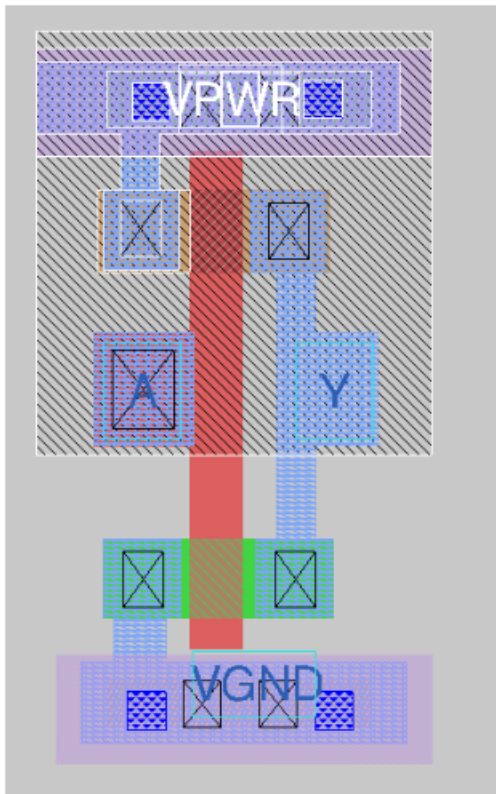
- **Objective:** Confirm that the drains of both MOSFETs form the inverter output.
 - To check connections, press s three times with the cursor on the OUT (Y) node:
 - First press: selects item under cursor (e.g., output net).
 - Second and third press: expands selection to all connected items.
 - The highlighted selection should include both NMOS and PMOS drains, confirming their connectivity at the output (Y node).



Verify Power and Ground Connections

- **Objective:** Ensure correct power architecture.
 - The **source of PMOS** must be connected to **VDD**.
 - The **source of NMOS** must be connected to **ground**.

- For PMOS:
 - Move the cursor over the VDD contact and press s to select it.
 - Press s again to expand selection to the entire region.
 - Confirm that the selected region correctly spans from the PMOS source to the VDD rail.



Reference: Building Inverter from Scratch

- An in-depth guide to building the inverter and further documentation is available on the referenced GitHub page.
- Review descriptions of **OpenLane** tools and the layout process, if additional information is required.

[Here](#)

8. Understand Abstract Cell Views (LEF)

- **Objective:** Grasp the abstraction provided by the LEF (Library Exchange Format) file.

- LEF describes **layout boundaries, metal layers, and pin locations** but omits **logic implementation** details for IP protection.
- Compare the detailed physical layout (showing transistors, wells, and polysilicon) with LEF (showing only cell edges and pin markers).
- In commercial tools, LEF may also be known as "frame view". LEF is essential for placement and routing, providing just the necessary physical cell information for automatic tools.

Lab steps to create std cell layout and extract spice netlist

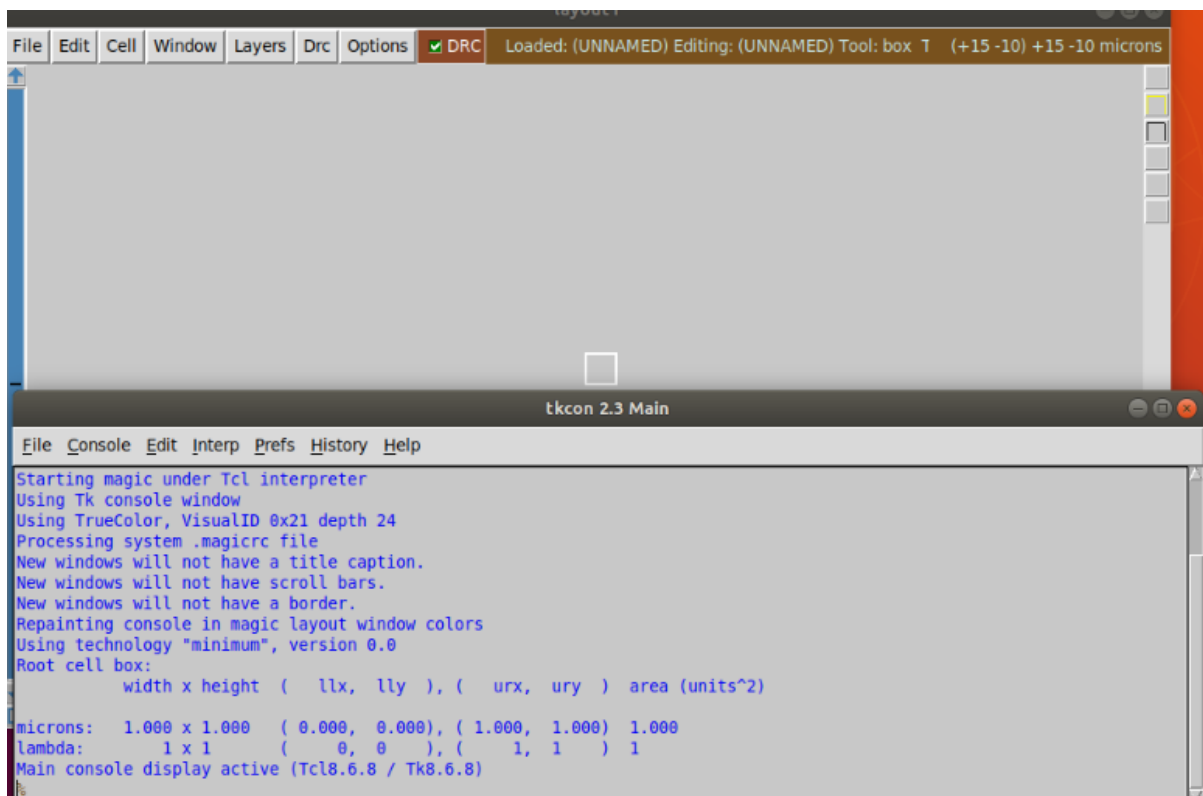
This lab demonstrates the **step-by-step creation of a standard cell** using the Magic VLSI layout tool. The aim is to define a fixed bounding box for the cell, lay down active layers (including wells, contacts, and metal interconnects), incrementally resolve design rule errors, and extract the cell for SPICE-level simulation.

The workflow includes defining precise coordinates, drawing and connecting all relevant layers (such as N-well, local interconnect, and metal 1), ensuring the cell is design-rule clean, and extracting the layout for simulation.

Lab Steps

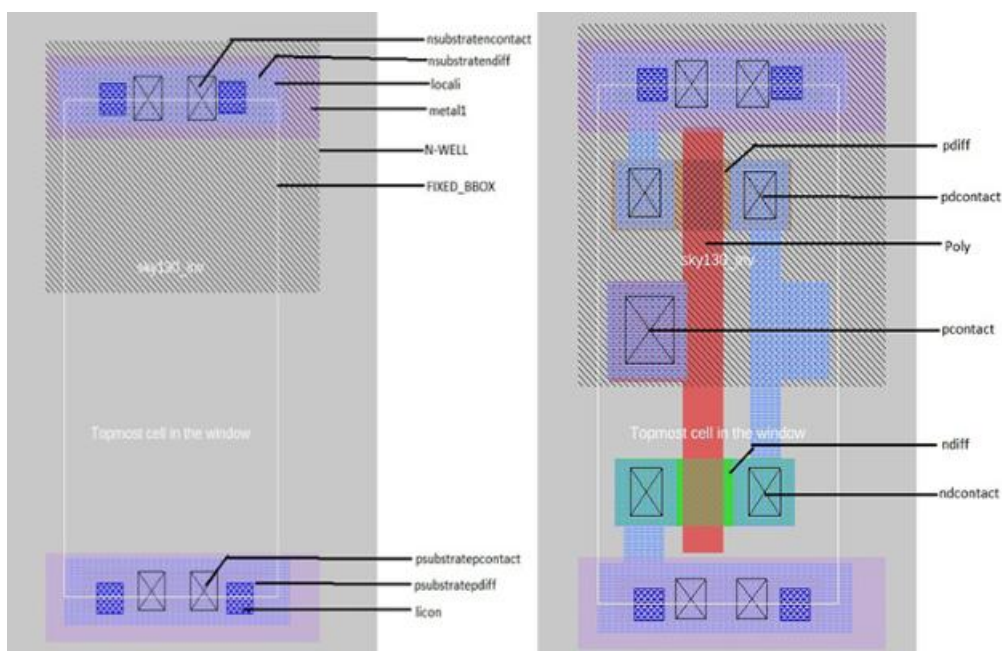
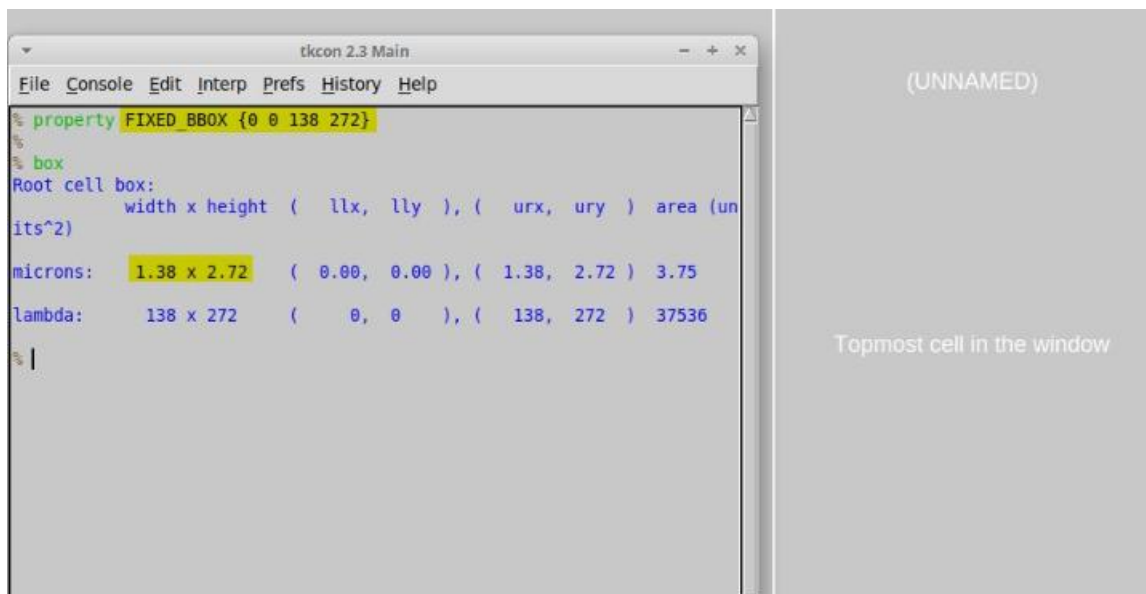
1. Start Magic and Open a New Layout

- Open Magic and ensure the layout window is empty for a fresh cell design.



Define the Fixed Bounding Box (Bbox)

- Create a fixed bounding box to set your standard cell's boundaries.
- Use command, specifying lower-left and upper-right coordinates (llx, lly, urx, ury):
- `box 0 0 138 272`
 - **Lower-left corner** is at (0,0).
 - **Upper-right corner** is at (138, 272) (unit: lambda, or 1.38μm by 2.72μm if using 0.01μm units).



Draw the N-well Region

- The dashed line represents the **N-well** for PMOS.
- Select the region corresponding to your PMOS and:
- paint nwell

4. Draw the Local Interconnect Layer (Poly or Locali)

- Use the paint tool to draw **Locali** (local interconnect) and connect between devices.
- paint locali

5. Draw Metal1 for Power and Ground Rails

- Standard cell rules require **power (VDD)** and **ground (GND)** rails to be on **Metal1**.
- paint metal1

6. Add Contacts Between Layers

a. N-well/Substrate Contact

- Needed to connect **nwell** and **locali** for PMOS.
- paint pcontact
 - Place where N-well and locali overlap.

b. Locali to Metal1 Contact (Licon)

- Needed to connect **locali** to **metal1**.
- paint licon
 - Place where locali and metal1 overlap.

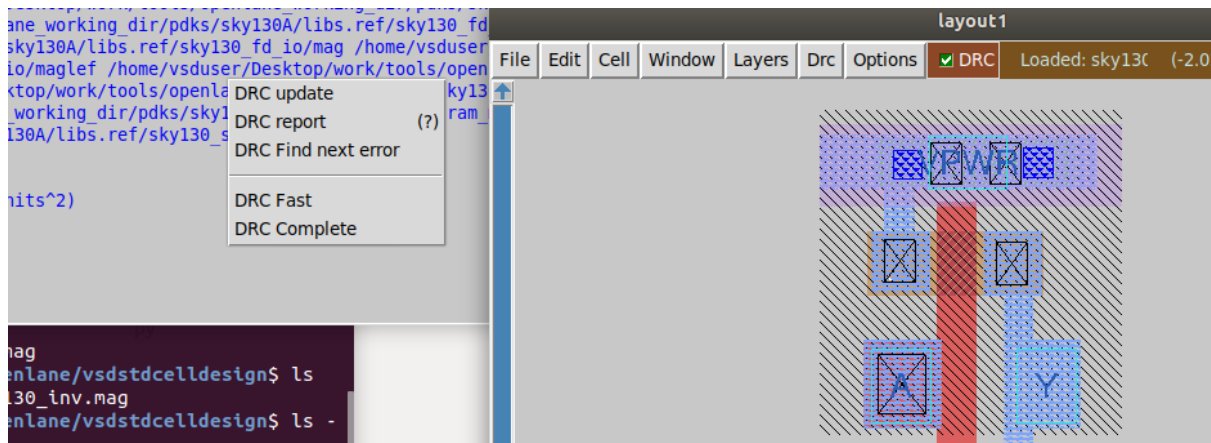
c. For NMOS: P-substrate to Locali Contact

- Draw contacts between the **p-substrate** (no explicit well) and **locali** for NMOS.
- paint ncontact

7. Check Layout Design Rules (DRC) Interactively

- Magic's DRC engine flags all errors during layout.
- To inspect errors:
 - Go to the DRC tab and click **Find Next Error**.
 - Magic will zoom in on DRC violations.

- Each violation is detailed in the terminal.



Ensure DRC Cleanliness

- The final design **must have zero DRC errors** for manufacturability.
- Confirm by checking that the DRC count is zero in the Magic interface.

9. Extract the Netlist for SPICE Simulation

- Open Magic's tkcon (Magic's TCL shell).
- Change directory to your working cell directory if needed.

a. Extract Layout to .ext File

```
```
```

```
pwd
```

```
extract all
```

```
```
```

- This creates an extraction file (e.g., sky_inv.ext).

```
tkcon 2.3 Main
File Console Edit Interp Prefs History Help

width x height ( llx, lly ), ( urx, ury ) area (units^2)
microns: 0.010 x 0.010 ( 0.000, 0.000), ( 0.010, 0.010) 0.000
lambda: 1 x 1 ( 0, 0 ), ( 1, 1 ) 1
main console display active (Tcl8.6.8 / Tk8.6.8)
s what
selected mask layers:
nmos ( Topmost cell in the window )
s what
selected mask layers:
pmos ( Topmost cell in the window )
s pwd
/home/vsduser/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign
s extract all
extracting sky130_inv into sky130_inv.ext:
```

```
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls
ltr
total 180
-rw-rw-r-- 1 vsduser vsduser 13525 Nov 11 21:13 README.md
-rw-rw-r-- 1 vsduser vsduser 11357 Nov 11 21:13 LICENSE
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 Images
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 extras
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vsduser vsduser 2716 Nov 11 21:13 sky130_inv.mag
-rwxr-xr-x 1 vsduser vsduser 136710 Nov 11 21:19 sky130A.tech
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$
```

View SPICE Netlist Contents

```

```
cat sky_inv.spice
```

```

- Inspect the generated netlist for correctness.

```

rect 23 264 40 281
rect 100 265 117 282
rect 21 -9 38 8
rect 105 -9 122 8
<< metal1 >>
rect -20 282 157 296
rect -20 281 100 282
rect -20 264 23 281
rect 40 265 100 281
rect 117 265 157 282
rect 40 264 157 265
rect -20 248 157 264
rect -11 8 157 24
rect -11 -9 21 8
rect 38 -9 105 8
rect 122 -9 157 8
rect -11 -24 157 -9
<< labels >>
flabel metal1 s 44 258 90 289 0 FreeSans 120 0 0 0 VPWR
port 3 nsew power bidirectional
flabel locali s 10 121 45 164 0 FreeSans 120 0 0 0 A
port 1 nsew signal input
flabel locali s 96 121 131 164 0 FreeSans 120 0 0 0 Y
port 2 nsew signal output
flabel metal1 s 50 -3 105 26 0 FreeSans 120 0 0 0 VGND
port 5 nsew ground bidirectional
<< properties >>
string LEFsite unithd
string LEFclass CORE
string FIXED_BBOX 0 0 138 272
string LEFsource USER
string LEForigin 0 0
<< end >>
vsduser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$

```

```

option scale=10m

subckt sky130_inv A Y VPWR VGND
0 Y A VGND VGND sky130_fd_pr__nfet_01v8 ad=1.44n pd=0.152m as=1.37n ps=0.148m w
35 l=23
1 Y A VPWR VPWR sky130_fd_pr__pfet_01v8 ad=1.44n pd=0.152m as=1.52n ps=0.156m w
37 l=23
0 VPWR Y 0.117f
1 A Y 0.0754f
2 A VPWR 0.0774f
3 Y VGND 0.279f
4 A VGND 0.45f
5 VPWR VGND 0.781f
ends

```

```

cd ..
vdsuser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls -
ltr
total 184
-rw-rw-r-- 1 vdsuser vdsuser 13525 Nov 11 21:13 README.md
-rw-rw-r-- 1 vdsuser vdsuser 11357 Nov 11 21:13 LICENSE
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 Images
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 extras
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vdsuser vdsuser 2716 Nov 11 21:13 sky130_inv.mag
-rwxr-xr-x 1 vdsuser vdsuser 136710 Nov 11 21:19 sky130A.tech
-rw-rw-r-- 1 vdsuser vdsuser 1365 Nov 11 21:44 sky130_inv.ext
vdsuser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls -
ltr
total 188
-rw-rw-r-- 1 vdsuser vdsuser 13525 Nov 11 21:13 README.md
-rw-rw-r-- 1 vdsuser vdsuser 11357 Nov 11 21:13 LICENSE
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 Images
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 extras
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vdsuser vdsuser 2716 Nov 11 21:13 sky130_inv.mag
-rwxr-xr-x 1 vdsuser vdsuser 136710 Nov 11 21:19 sky130A.tech
-rw-rw-r-- 1 vdsuser vdsuser 1365 Nov 11 21:44 sky130_inv.ext
-rw-rw-r-- 1 vdsuser vdsuser 404 Nov 11 21:46 sky130_inv.spice
vdsuser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$

```

File Console Edit Interp Prefs History Help

```

c:/tools/openlane_working_dir/pdks/sky130A/libs.ref/sky130_ml_xx_hd/mag /home/vdsuser/Desktop/work/t
working_dir/pdks/sky130A/libs.ref/sky130_sram_macros/mag /home/vdsuser/Desktop/work/tools/openlane_v
/sky130A/libs.ref/sky130_sram_macros/maglef"

```

Using technology "sky130A", version 1.0.141-0-gb9ffc1f

root cell box:

width x height (llx, lly), (urx, ury) area (units^2)

microns: 0.010 x 0.010 (0.000, 0.000), (0.010, 0.010) 0.000

lambda: 1 x 1 (0, 0), (1, 1) 1

Main console display active (Tcl8.6.8 / Tk8.6.8)

% extract all

Extracting sky130_inv into sky130_inv.ext:

% ext2spice cthresh 0 rthresh 0

% ext2spice

ext2spice finished.

% |

```

rwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 extras
rwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vdsuser vdsuser 2716 Nov 11 21:13 sky130_inv.mag
-rwxr-xr-x 1 vdsuser vdsuser 136710 Nov 11 21:19 sky130A.tech
-rw-rw-r-- 1 vdsuser vdsuser 1365 Nov 11 21:44 sky130_inv.ext
vdsuser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls -
ltr
total 188
-rw-rw-r-- 1 vdsuser vdsuser 13525 Nov 11 21:13 README.md
-rw-rw-r-- 1 vdsuser vdsuser 11357 Nov 11 21:13 LICENSE
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 Images
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 extras
drwxrwxr-x 2 vdsuser vdsuser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vdsuser vdsuser 2716 Nov 11 21:13 sky130_inv.mag
-rwxr-xr-x 1 vdsuser vdsuser 136710 Nov 11 21:19 sky130A.tech
-rw-rw-r-- 1 vdsuser vdsuser 1365 Nov 11 21:44 sky130_inv.ext
-rw-rw-r-- 1 vdsuser vdsuser 404 Nov 11 21:46 sky130_inv.spice
vdsuser@raheem:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$

```

```

-rw-rw-r-- 1 vsduser vsduser 1365 Nov 11 21:44 sky130_inv.ext
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ ls -
ltr
total 188
-rw-rw-r-- 1 vsduser vsduser 13525 Nov 11 21:13 README.md
-rw-rw-r-- 1 vsduser vsduser 11357 Nov 11 21:13 LICENSE
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 Images
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 extras
drwxrwxr-x 2 vsduser vsduser 4096 Nov 11 21:13 libs
-rw-rw-r-- 1 vsduser vsduser 2716 Nov 11 21:13 sky130_inv.mag
-rwxr-xr-x 1 vsduser vsduser 136710 Nov 11 21:19 sky130A.tech
-rw-rw-r-- 1 vsduser vsduser 1365 Nov 11 21:44 sky130_inv.ext
-rw-rw-r-- 1 vsduser vsduser 404 Nov 11 21:46 sky130_inv.spice
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$ cat
sky130_inv.spice
* SPICE3 file created from sky130_inv.ext - technology: sky130A

.option scale=10m

.subckt sky130_inv A Y VPWR VGND
X0 Y A VGND VGND sky130_fd_pr__nfet_01v8 ad=1.44n pd=0.152m as=1.37n ps=0.148m w=35 l=23
X1 Y A VPWR VPWR sky130_fd_pr__pfet_01v8 ad=1.44n pd=0.152m as=1.52n ps=0.156m w=37 l=23
C0 Y A 0.0754f
C1 VPWR A 0.0774f
C2 Y VPWR 0.117f
C3 Y VGND 0.279f
C4 A VGND 0.45f
C5 VPWR VGND 0.781f
.ends
vsduser@raheen:~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign$

```

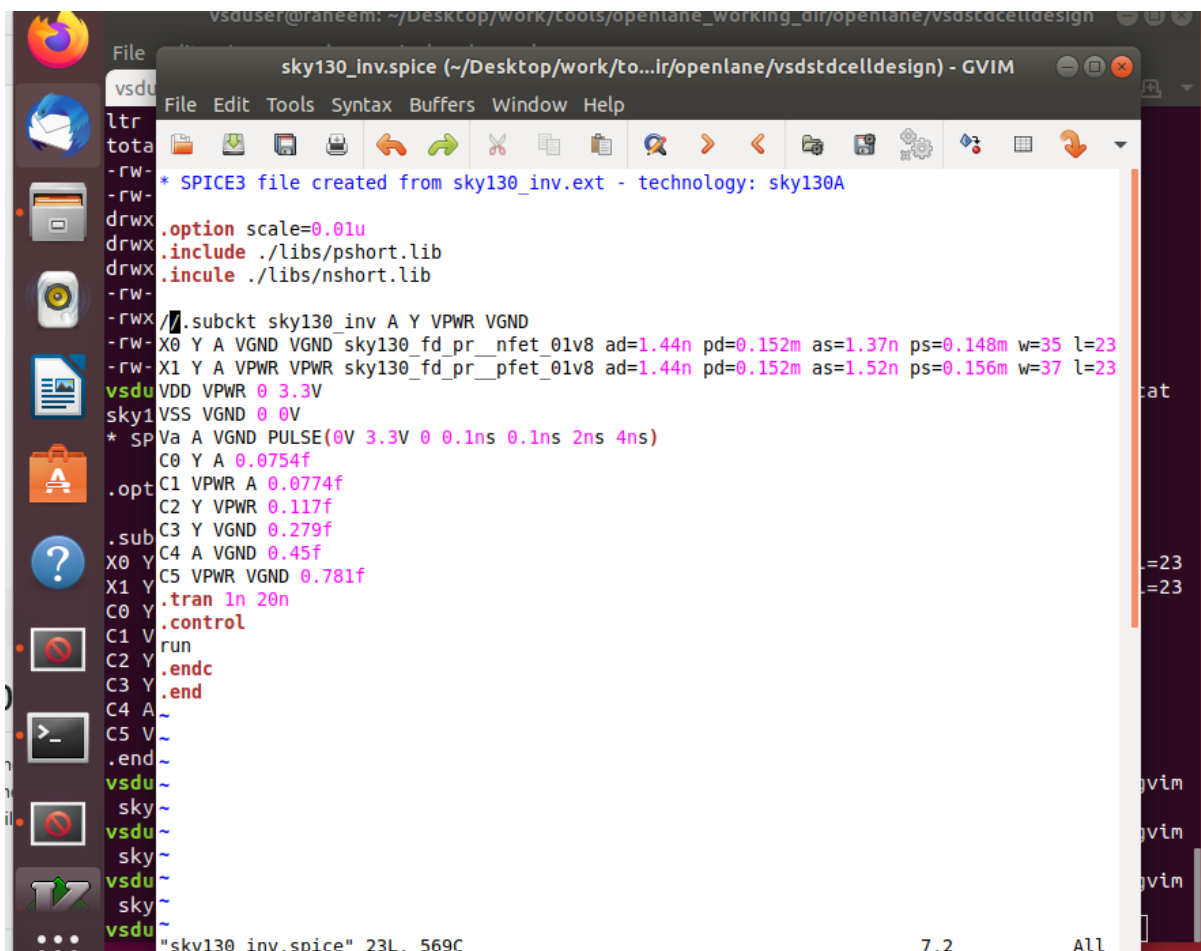
```

sky130_inv.spice + (~/Desktop/work/.../openlane/vsdstdcelldesign) - GVIM
File Edit Tools Syntax Buffers Window Help
[Icons]
* SPICE3 file created from sky130_inv.ext - technology: sky130A

.option scale=0.01u
.include ./libs/pshort.lib
.incule ./libs/nshort.lib

.subckt sky130_inv A Y VPWR VGND
<0 Y A VGND VGND sky130_fd_pr__nfet_01v8 ad=1.44n pd=0.152m as=1.37n ps=0.148m w=35 l=2
<1 Y A VPWR VPWR sky130_fd_pr__pfet_01v8 ad=1.44n pd=0.152m as=1.52n ps=0.156m w=37 l=2
:0 Y A 0.0754f
:1 VPWR A 0.0774f
:2 Y VPWR 0.117f
:3 Y VGND 0.279f
:4 A VGND 0.45f
:5 VPWR VGND 0.781f
.ends

```



```
sky130_inv.spice (~/Desktop/work/to...ir/openlane/vsdstdcclldesign) - GVIM
File Edit Tools Syntax Buffers Window Help

* SPICE3 file created from sky130_inv.ext - technology: sky130A

.option scale=0.01u
.include ./libs/pshort.lib
.incule ./libs/nshort.lib

.subckt sky130_inv A Y VPWR VGND
X0 Y A VGND VGND sky130_fd_pr_nfet_01v8 ad=1.44n pd=0.152m as=1.37n ps=0.148m w=35 l=23
-rw- X1 Y A VPWR VPWR sky130_fd_pr_pfet_01v8 ad=1.44n pd=0.152m as=1.52n ps=0.156m w=37 l=23
-rw- VDD VPWR 0 3.3V
sky1 VSS VGND 0 0V
* SP Va A VGND PULSE(0V 3.3V 0 0.1ns 0.1ns 2ns 4ns)
C0 Y A 0.0754f
C1 VPWR A 0.0774f
C2 Y VPWR 0.117f
C3 Y VGND 0.279f
C4 A VGND 0.45f
C5 VPWR VGND 0.781f
.tran 1n 20n
.control
run
.endc
.end

vsdu ~
sky ~
sky ~
vsdu ~
sky ~
vsdu ~

"sky130_inv.spice" 23L, 569C 7.2 All
```

Lab steps to characterize inverter using sky130 model files

This lab focuses on characterizing a standard inverter cell by analyzing its transient response and extracting key performance metrics such as rise/fall transition times and propagation delays ("cell delays") at room temperature. The steps include running SPICE simulations, plotting time-domain responses, adjusting load capacitance, and conducting detailed measurements on simulation results. All steps are illustrated with plot interactions and calculations.

Lab Steps

1. Run Initial SPICE Simulation and Plot Output

```
ngspice sky130_inv.spice
```

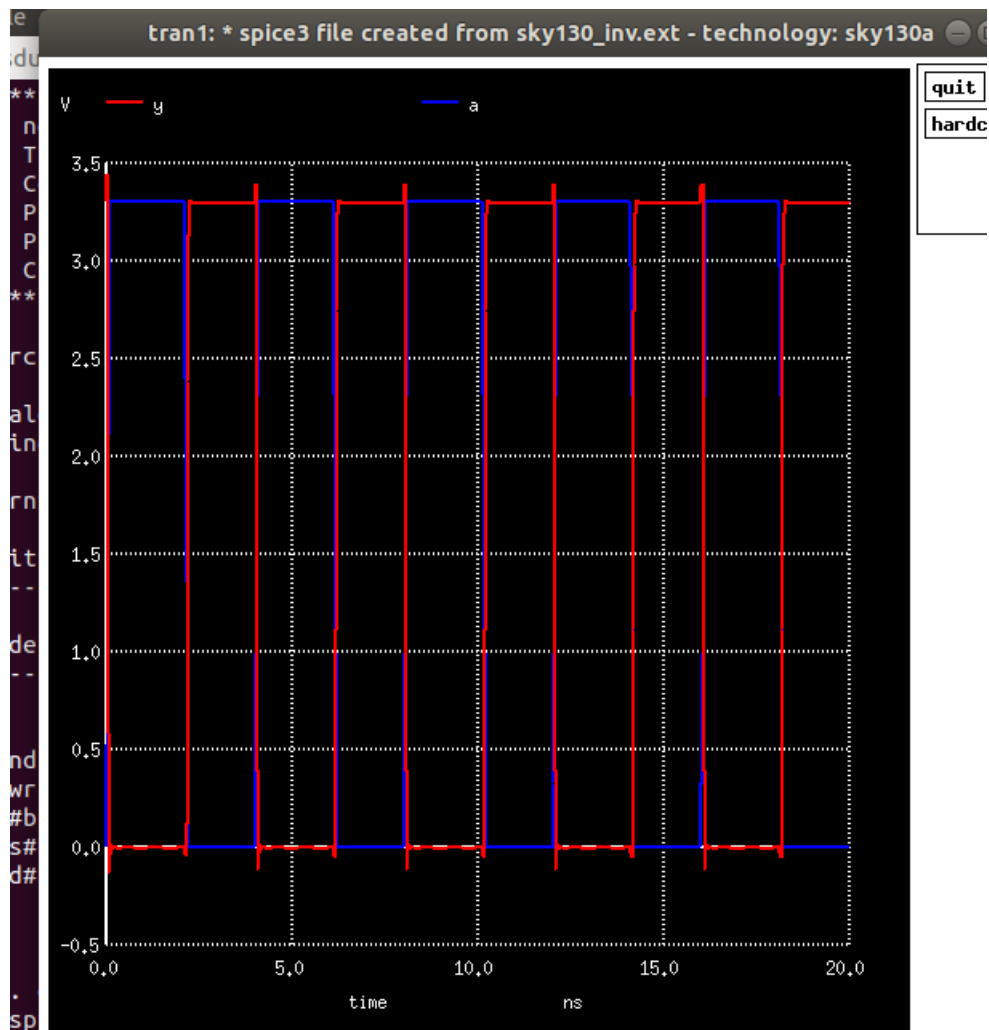
```
sky130_inv.spice + (~/Desktop/work/...ir/openlane/vsdstdcelldesign) - GVIM
File Edit Tools Syntax Buffers Window Help
* SPICE3 file created from sky130_inv.ext - technology: sky130A
.option scale=0.01u
.include ./libs/pshort.lib
.incule ./libs/nshort.lib

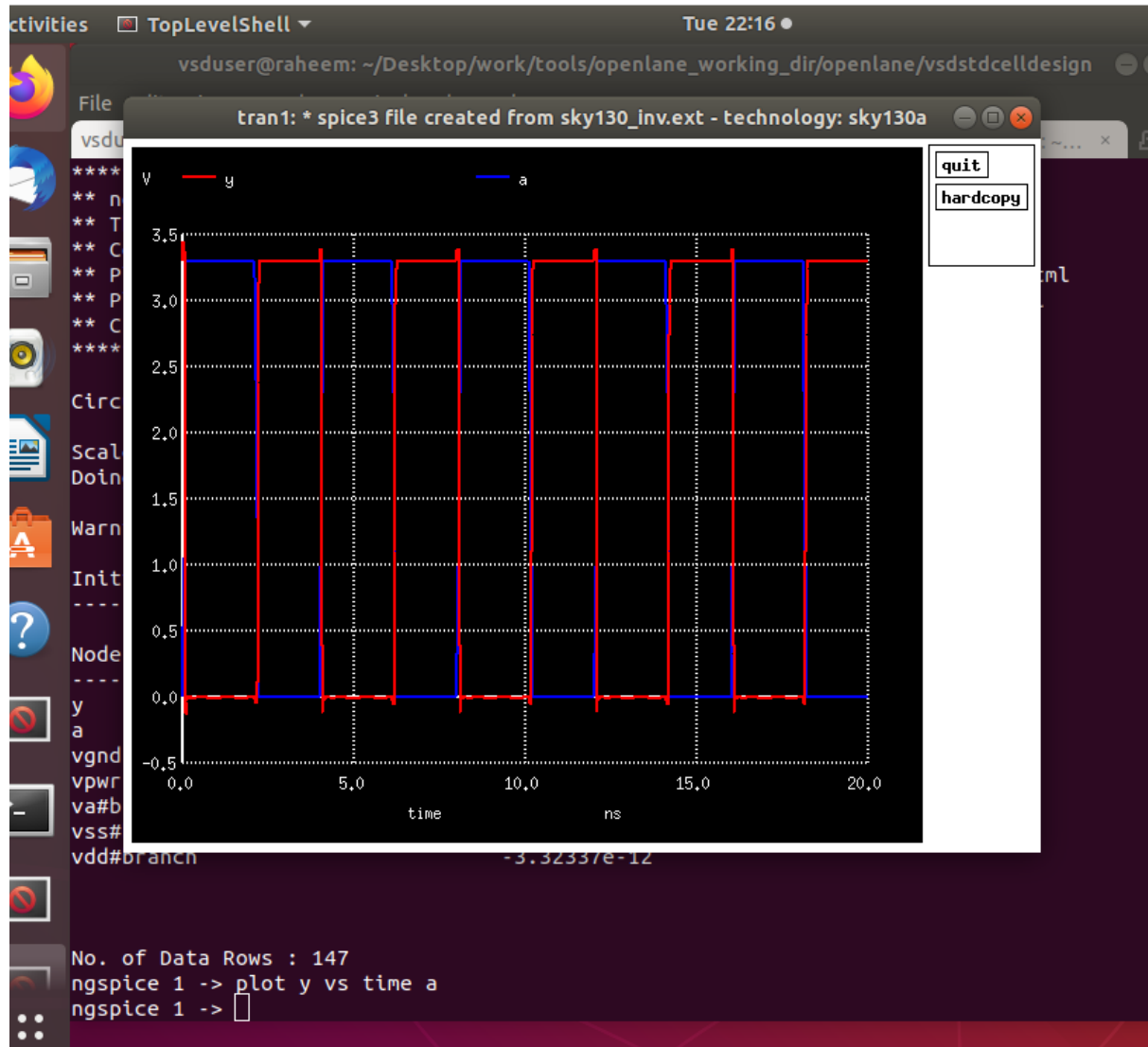
//.subckt sky130_inv A Y VPWR VGND
M1001 Y A VGND VGND sky130_fd_pr__nfet_01v8 ad=1.44n pd=0.152m as=1.37n ps=0.148m w=35 l
=23
M1000 Y A VPWR VPWR sky130_fd_pr__pfet_01v8 ad=1.44n pd=0.152m as=1.52n ps=0.156m w=37 l
=23
VDD VPWR 0 3.3V
VSS VGND 0 0V
Va A VGND PULSE(0V 3.3V 0 0.1ns 0.1ns 2ns 4ns)
C0 Y A 0.0754f
C1 VPWR A 0.0774f
C2 Y VPWR 0.117f
C3 Y VGND 0.279f
C4 A VGND 0.45f
C5 VPWR VGND 0.781f
.tran 1n 20n
.control
run
.endc
.end
~
~
```

```
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign
File Edit View Search Terminal Tabs Help
vsduser@raheem: ~... x vsduser@raheem: ~... x vsduser@raheem: ~... x vsduser@raheem: ~... x
Trying gmin = 1.0000E-10 Note: One successful gmin step
Trying gmin = 1.0000E-11 Note: One successful gmin step
Trying gmin = 1.0000E-12 Note: One successful gmin step
Note: One successful source step
Supplies reduced to 0.1000% Warning: singular matrix: check nodes y and y
Supplies reduced to 0.0000% Warning: singular matrix: check nodes y and y
Warning: source stepping failed
Transient solution failed -
Last Node Voltages
-----
Node                               Last Voltage          Previous Iter
----                               -
y                                  0                     0
a                                  0                     0
vgnd                               0                     0
sky130_fd_pr__nfet_01v8           0                     0
ad                                  0                     0
1.44n                              0                     0
vpwr                               0                     0
sky130_fd_pr__pfet_01v8           0                     0
va#branch                         0                     0
vss#branch                        0                     0
vdd#branch                        0                     0
doAnalyses: iteration limit reached
run simulation(s) aborted
ngspice 1 -> 
```

```
vsdu File Edit Tools Syntax Buffers Window Help
vsdu * SPICE3 file created from sky130_inv.ext - technology: sky130A
extr
Imag
vsdu
sky .option scale=0.01u
vsdu .include ./libs/pshort.lib
sky .includ ./libs/nshort.lib
vsdu
sky // .subckt sky130_inv A Y VPWR VGND
vsdu M1001 Y A VGND VGND nshort_model.0 ad=1.44n pd=0.152m as=1.37n ps=0.148m w=35 l=23
sky M1000 Y A VPWR VPWR pshort_model.0 ad=1.44n pd=0.152m as=1.52n ps=0.156m w=37 l=23
vsdu VDD VPWR 0 3.3V
VSS VGND 0 0V
Va A VGND PULSE(0V 3.3V 0 0.1ns 0.1ns 2ns 4ns)
C0 Y A 0.0754f
C1 VPWR A 0.0774f
C2 Y VPWR 0.117f
C3 Y VGND 0.279f
C4 A VGND 0.45f
C5 VPWR VGND 0.781f
.tran 1n 20n
.control
run
.endc
.end
~
~
~
~
```

```
vsduser@raheem: ~/Desktop/work/tools/openlane_working_dir/openlane/vsdstdcelldesign
File Edit View Search Terminal Tabs Help
vsduser@raheem: ~... x vsduser@raheem: ~... x vsduser@raheem: ~... x vsduser@raheem: ~... x
ice sky130_inv.spice
*****
** ngspice-27 : Circuit level simulation program
** The U. C. Berkeley CAD Group
** Copyright 1985-1994, Regents of the University of California.
** Please get your ngspice manual from http://ngspice.sourceforge.net/docs.html
** Please file your bug-reports at http://ngspice.sourceforge.net/bugrep.html
** Creation Date: Tue Dec 26 17:10:20 UTC 2017
*****
Circuit: * spice3 file created from sky130_inv.ext - technology: sky130a
Scale set
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
Warning: va: no DC value, transient time 0 value used
Initial Transient Solution
-----
Node Voltage
----
y 3.3
a 0
vgnd 0
vpwr 3.3
va#branch 0
vss#branch 3.32336e-12
vdd#branch -3.32337e-12
No. of Data Rows : 147
ngspice 1 ->
```





```

* SPICE3 file created from sky130_inv.ext - technology: sky130A

.option scale=0.01u
.include ./libs/pshort.lib
.include ./libs/nshort.lib

//.subckt sky130_inv A Y VPWR VGND
M1001 Y A VGND VGND nshort_model.0 ad=1.44n pd=0.152m as=1.37n ps=0.148m w=35 l=
23
M1000 Y A VPWR VPWR pshort_model.0 ad=1.44n pd=0.152m as=1.52n ps=0.156m w=37 l=
23
VDD VPWR 0 3.3V
VSS VGND 0 0V
Va A VGND PULSE(0V 3.3V 0 0.1ns 0.1ns 2ns 4ns)
C0 Y A 0.0754f
C1 Y VPWR 0.117f
C2 VPWR A 0.0774f
C3 Y VGND 0.279f
C4 A VGND 0.45f
C5 VPWR VGND 0.781f
//.ends
.tran 1n 20n

.control
run
.endc
.end

```

sky130_inv.spice (~/Desktop/work...openlane/vsdstdcell/design) - GVIM

File Edit Tools Syntax Buffers Window Help



```

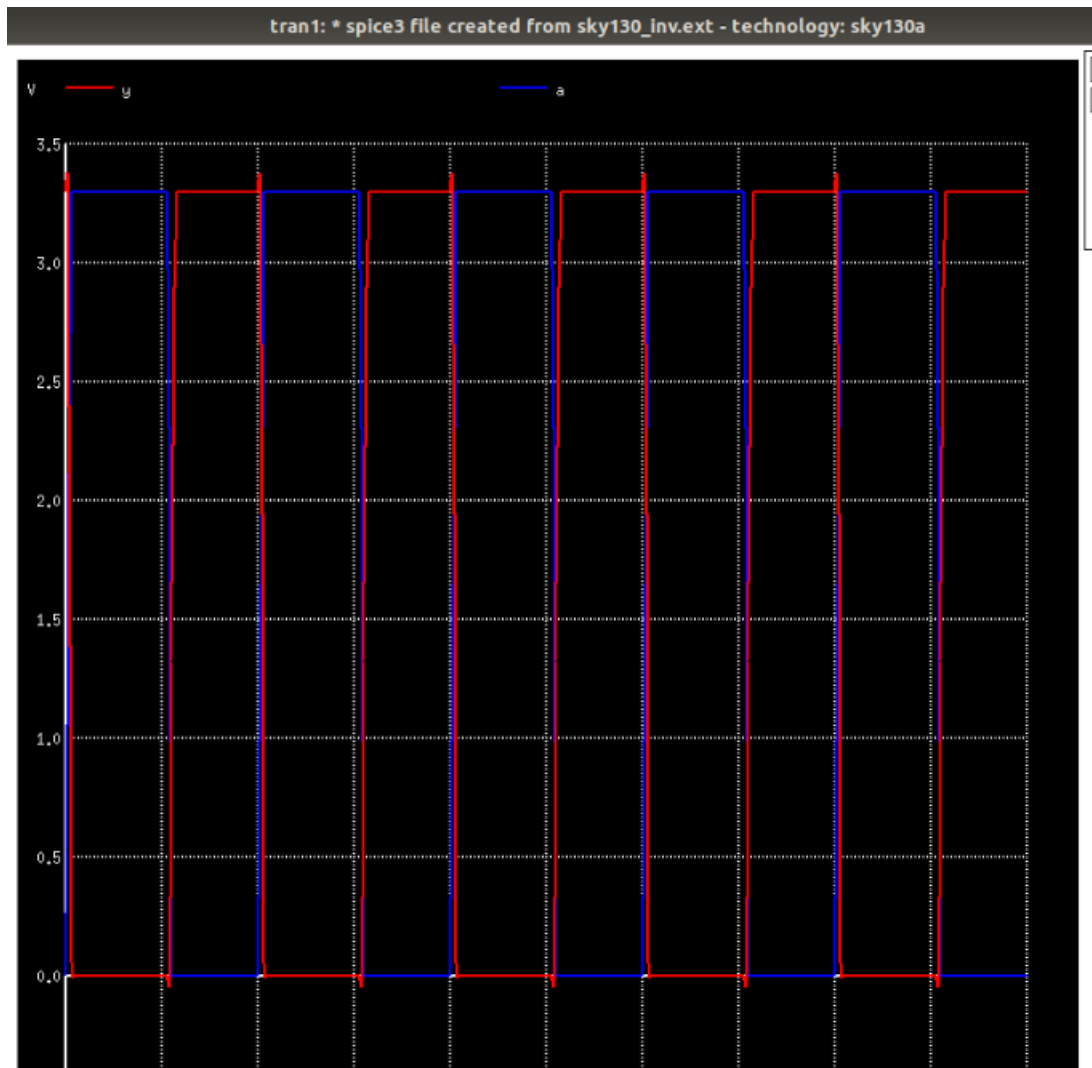
* SPICE3 file created from sky130_inv.ext - technology: sky130A

.option scale=0.01u
.include ./libs/pshort.lib
.include ./libs/nshort.lib

//.subckt sky130_inv A Y VPWR VGND
M1001 Y A VGND VGND nshort_model.0 ad=1.44n pd=0.152m as=1.37n ps=0.148m w=35 l=
23
M1000 Y A VPWR VPWR pshort_model.0 ad=1.44n pd=0.152m as=1.52n ps=0.156m w=37 l=
23
VDD VPWR 0 3.3V
VSS VGND 0 0V
Va A VGND PULSE(0V 3.3V 0 0.1ns 0.1ns 2ns 4ns)
C0 Y A 0.0754f
C1 Y VPWR 0.117f
C2 VPWR A 0.0774f
C3 Y VGND 2f
C4 A VGND 0.45f
C5 VPWR VGND 0.781f
//.ends
.tran 1n 20n

.control

```



Plot New Transient Output

- Plot the **output voltage versus time**, with the adjusted load capacitance.
- Observe the output for improved waveforms.

4. Identify and Measure Rise Transition Time

- The **rise transition** is defined as the time required for the output to transition from **20% to 80%** of its maximum value.
- Given the supply voltage $V_{DD} = 3.3V$:
 - 20 of $3.3V$ is $0.66V$.
 - 80 of $3.3V$ is $2.64V$.
- Use the plot:
 - Zoom in near the output rise edge.
 - Locate the time at which the output voltage first crosses $0.66V$ (let's say

Extra open brace or missing close brace

$$t_{20\%} = 2.18\text{ns}$$

).

- Locate the time at which the output voltage reaches 2.64V (let's say

Extra open brace or missing close brace

$$t_{80\%} = 2.245\text{ns}$$

Measure Fall Transition Time

- The **fall transition** is the time required for the output to fall from **80% to 20%** (2.64V down to 0.66V).
- Follow the same procedure as above by examining the output fall region.
- The value is to be determined as part of your assignment.

6. Measure Cell Propagation (Rise) Delay

- The **cell rise delay** is the time difference between the 50 point of the input transition and the 50 point of the output transition on a rising edge.
- Calculate 50 of 3.3V : 1.65V .
- On the plot:
 - Find the time when the **output** rises through 1.65V (e.g.,

Extra open brace or missing close brace

$$t_{\text{out},50\%} = 2.211\text{ns}$$

).

- Find the corresponding time when the **input** crosses 1.65V on its rising edge (e.g.,

Extra open brace or missing close brace

$$t_{in,50\%} = 2.150, ns$$

).

- Compute **cell rise delay**:
 - Extra open brace or missing close brace

$$t_{rise, delay} = t_{out, 50\%} - t_{in, 50\%} = 2.211, ns - 2.150, ns = 0.061, ns$$

7. Measure Cell Fall Delay

- The **cell fall delay** is the time difference between the 50 point of a falling output and the 50 point of the input transition.
- This step will be completed as part of your assignment.

8. Note the Simulation Temperature

- The characterization was performed at a temperature of **27°C** (room temperature).

```
Circuit: * spice3 file created from sky130_inv.ext - technology: sky130a
Scale set
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```